Subexponential Set Splitting through Selector Moments and Exact #3SAT

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Abstract

We give a variable-efficient reduction from the complete split-count distribution of an unbounded-rank hypergraph to exact counting for 3-CNF formulas. For a hypergraph with n vertices and m indexed edges, the reduction makes $m+1$ counting calls, each with $O(n+m)$ variables and $O(\sum_e |e| + m + 1)$ clauses. Selectors mark arbitrary subsets of monochromatic edges; fixing the number of selectors computes binomial moments, from which the whole distribution is recovered exactly. Combining this reduction with the general-density exact-counting theorem stated in the October 2026 SAT manuscript gives deterministic polynomial-space time

$$2^{O(k \log\log k/\sqrt{\log k})} + |I|^{O(1)}$$

for Set Splitting parameterized by the target number k of split sets. Using only the weaker rate documented in the accompanying formalization scope note gives $2^{O(k(\log\log k)^2/\sqrt{\log k})} + |I|^{O(1)}$. The same bounds, with k replaced by $n+m$, compute the original hypergraph's full distribution and an optimal coloring. The reduction also applies to counting the number of satisfied predicates in a family of 2-CNF formulas, and hence to the complete satisfaction profile of arbitrary-width CNF. All subexponential bounds explicitly depend on the imported counting theorem; the reductions and counting identities do not.

1 Introduction

An edge of a hypergraph is *split* by a Boolean coloring if it contains both colors. In the parameterized Set Splitting problem, the input is a hypergraph and a target k, and the question is whether some coloring splits at least k edges. The parameter counts edges that must be split; it does not bound their cardinality. We allow arbitrary edge sizes.

Lokshtanov and Saurabh [1] give a polynomial-time decision kernel with at most k vertices and $2k$ edges. Specialized algorithms of Nederlof, van Rooij, and van Dijk [2] achieve $O^*(1.7171^k)$ time with exponential space and $O^*(1.8213^k)$ time with polynomial space. Here O^* suppresses polynomial factors. These are useful established reference points for the present transfer result.

The recent manuscript *The Exponential Time Hypothesis is False* states subexponential exact-counting algorithms for each fixed CNF width [3]. A direct encoding of whether every hyperedge splits uses clauses whose width equals the edge size. Replacing large OR gates by bounded-fan-in circuits can introduce one auxiliary variable per incidence. Even after the decision kernel, this can produce $\Theta(k^2)$ variables, which does not by itself give a subexponential dependence on k from the imported quantitative bound.

We avoid this expansion by counting selected subsets of *unsplit* edges. A selector need only imply monochromaticity. Monochromaticity is a conjunction of pairwise equalities, so each guarded condition is a clause of width at most three without an incidence variable. A cardinality circuit on the selectors uses only $O(m)$ additional variables. Its counts are binomial moments of the unsplit-edge distribution, which can be inverted in polynomial time.

Contributions. The main unconditional reduction, Theorem 3.3, uses $O(n+m)$ variables regardless of edge rank. Theorem 4.1 transfers the two documented general-density counting rates to full split profiles. Corollary 4.2 combines the transfer with the decision kernel. Section 5 proves extensions to pinned colors, optimal colorings, and families of 2-CNF predicates.

External dependence and status. We treat the quantitative counting statements in Section 2 as explicit algorithmic premises. This paper neither reproduces their proofs nor independently certifies their formalization. The stronger manuscript rate and the weaker rate documented in its formalization scope note are kept separate. This is a research draft, not a claim of external review or publication priority. The proposed contribution is the quantitative, variable-efficient application of established counting ideas to unbounded-rank split profiles.

2 Preliminaries and counting premises

A hypergraph is $H = (V, (e_i)_{i=1}^m)$, with $V = [n]$ and $e_i \subseteq V$. Edges are indexed: parallel edges remain distinct occurrences. Put

$$s = n + m, \qquad J = \sum_{i=1}^{m} |e_i|.$$

The universe and incidence lists are explicit. Let L denote their bit encoding length, including isolated vertices and edge labels; polynomial factors may depend on $L+1$. Empty and singleton edges are never split. For convenience, we call them monochromatic, including the vacuous empty-edge case.

For $x \in \{0,1\}^n$, let $u_H(x)$ be the number of monochromatic edges and $b_H(x) = m - u_H(x)$ the number of split edges. Define

$$a_j(H) = |\{x \in \{0,1\}^n : u_H(x) = j\}|, \tag{1}$$

$$Q_H(z) = \sum_{j=0}^{m} a_j(H) z^j, \qquad P_H(z) = \sum_{x \in \{0,1\}^n} z^{b_H(x)} = \sum_{j=0}^{m} a_j(H) z^{m-j}. \tag{2}$$

Thus P_H is the *split-count polynomial*. For nonempty edges, Q_H is a specialization of the hypergraph coboundary polynomial studied by White [5]. Neither this polynomial nor its usual deletion–contraction identity is claimed to be new.

All logarithmic expressions below are for sufficiently large arguments; finitely many small cases are solved directly. For $\alpha \in \{1,2\}$, write

$$f_\alpha(v) = \frac{v(\log\log v)^\alpha}{\sqrt{\log v}}.$$

The choice of logarithm base changes only constants. A monotone extension handles small arguments.

Definition 2.1 (Counting interfaces)**.** Interface Count_α is a uniform deterministic algorithm that exactly counts satisfying assignments of a 3-CNF with v declared variables and encoding length ℓ in time

$$(\ell+1)^{O(1)}2^{O(f_\alpha(v))}$$

and space polynomial in $\ell + v + 1$. Interface $\mathsf{Count}_{\mathrm{sp}}$ has, for every fixed D, time $\mathrm{poly}(\ell + v)2^{O_D(v/\sqrt{\log v})}$ on formulas with at most Dv clauses, and polynomial space.

The revised October 25 manuscript states Count_1 and $\mathsf{Count}_{\mathrm{sp}}$ as specializations of its fixed-width main theorem [3]. The accompanying scope note documents Count_2 [4]. We use width three throughout; no hidden constant is allowed to depend on a growing edge size. The interfaces count assignments to *declared* variables, including isolated ones. If an implementation drops such variables, their factor of two must be restored. Unit and binary clauses can be padded by repeating literals when an interface requires exactly three literal occurrences.

3 Selector moments and a linear-variable encoding

Define the binomial moments

$$F_t(H) = \sum_{x\in\{0,1\}^n} \binom{u_H(x)}{t}, \qquad 0 \le t \le m, \tag{3}$$

where $\binom{j}{t} = 0$ for $t > j$. The selectors below count subsets of monochromatic edges rather than requiring every monochromatic edge to be selected.

Lemma 3.1 (Binomial inversion)**.** *The moments determine the whole profile by*

$$a_j(H) = \sum_{t=j}^{m} (-1)^{t-j}\binom{t}{j} F_t(H). \tag{4}$$

Equivalently,

$$P_H(z) = \sum_{t=0}^{m} F_t(H) z^{m-t}(1-z)^t. \tag{5}$$

The transformation uses polynomially many bit operations in $n + m + 1$.

Proof. Let $B_H(w) = \sum_{t=0}^{m} F_t(H)w^t$. The binomial theorem gives

$$B_H(w) = \sum_x (1+w)^{u_H(x)} = Q_H(1+w).$$

Hence $Q_H(z) = B_H(z-1)$; taking coefficients proves (4). Substituting z^{-1} and multiplying by z^m gives the polynomial identity (5). Each $F_t \le 2^{n+m}$. Every term of (4) has magnitude at most 2^{n+2m}, and each partial sum has magnitude at most $(m+1)2^{n+2m}$. Thus $O(n+m+\log(m+1))$ bits suffice, including intermediate signed arithmetic. An $O(m^2)$-term implementation is polynomial in the explicit output size. □

Lemma 3.2 (Linear-size exact cardinality circuit)**.** *For m Boolean inputs and $0 \le t \le m$, the predicate that exactly t inputs equal one has a fan-in-two Boolean circuit with $O(m+1)$ gates. It admits an equivalent width-three Tseitin encoding with $O(m+1)$ auxiliary variables and clauses, and a unique auxiliary extension for every satisfying input assignment.*

Proof. Pad the input list with zeros to a power of two less than $2m$ when $m > 0$. Sum the bits in a balanced binary tree. At level h, there are $O(m/2^h)$ additions on $O(h)$-bit operands, each requiring $O(h)$ bounded-fan-in gates. The total is

$$O\left(m \sum_{h \geq 1} \frac{h}{2^h}\right) = O(m).$$

Fixing the output bits to the binary expansion of t uses $O(\log(m+1))$ unit conditions. Constants and $m = 0$ require only constant overhead.

Introduce a variable for each gate and encode its full logical equivalence. For example, $g \leftrightarrow (a \wedge b)$ is encoded by

$$(\neg g \vee a) \ \wedge \ (\neg g \vee b) \ \wedge \ (g \vee \neg a \vee \neg b).$$

OR, NOT, and a constant-size expansion of XOR have width-three encodings. Once the input bits are fixed, each gate variable is uniquely determined in topological order. Consequently satisfying inputs have exactly one extension, while nonsatisfying inputs have none. □

Theorem 3.3 (Selector encoding)**.** *For every hypergraph H and $0 \leq t \leq m$, one can construct in polynomial time a 3-CNF $\Psi_{H,t}$ with*

$$v = O(n + m + 1), \qquad c = O(J + m + 1)$$

variables and clauses, respectively, such that $\#\Psi_{H,t} = F_t(H)$. The implicit constants are absolute and independent of edge rank. Its bit encoding length is $O((J + n + m + 1)\log(n + m + 2))$.

Proof. Use the color variables $x_1, \ldots, x_n$ and introduce a selector y_i for each indexed edge e_i. For a nonempty e_i, choose an anchor $a_i \in e_i$. For every $v \in e_i \setminus \{a_i\}$ include the clauses

$$(\neg y_i \vee \neg x_{a_i} \vee x_v), \qquad (\neg y_i \vee x_{a_i} \vee \neg x_v). \tag{6}$$

They express $y_i \Rightarrow (x_{a_i} = x_v)$. No extra variable is introduced for an incidence. Empty and singleton edges need no guarded clauses, because they are always unsplit. Finally add the exact-t circuit and the unique-extension encoding of Lemma 3.2 on the m selectors.

Fix a coloring x. A selector may equal one only on a monochromatic edge; on each such edge it may independently equal zero or one. The exact-t condition therefore permits exactly $\binom{u_H(x)}{t}$ selector assignments. For every permitted selector assignment the gate variables have one extension. Summing over x proves the count identity.

The guarded clauses number $2\sum_i \max\{|e_i| - 1, 0\} \leq 2J$. The cardinality construction adds $O(m+1)$ variables and clauses. The declared color variables contribute n, even if they are isolated. All variable indices use $O(\log(n + m + 2))$ bits, giving the asserted length. □

Remark 3.4 (The one-way implication is essential)**.** The guards are deliberately *not* equivalences between y_i and monochromaticity. Requiring equivalence would destroy the subset-counting interpretation. In contrast, the auxiliary *gate* constraints must be equivalences to obtain a unique gate extension. These two roles are different.

3.1 A small example

Let $V = \{a, b, c\}$ and let the edges be $\{a, b\}$ and $\{b, c\}$. There are two colorings with no monochromatic edge, four with one, and two with two. Thus

$$(a_0, a_1, a_2) = (2, 4, 2), \qquad (F_0, F_1, F_2) = (8, 8, 2).$$

Inversion returns $a_2 = 2$, $a_1 = 8 - 2 \cdot 2 = 4$, and $a_0 = 8 - 8 + 2 = 2$. The resulting split-count polynomial is $P_H(z) = 2 + 4z + 2z^2$. This example illustrates why counting only the existence of a selector assignment would lose the needed information.

4 Subexponential algorithms

Theorem 4.1 (Complete split profiles)**.** *Assume* Count_α *for* $\alpha \in \{1, 2\}$*. On every explicitly represented hypergraph with* $s = n + m$*, all coefficients of* P_H *can be computed deterministically in polynomial space and time*

$$(L + 1)^{O(1)} 2^{O(f_\alpha(s))}. \tag{7}$$

There is no restriction on rank, parallel edges, or isolated vertices.

Proof. For $t = 0, 1, \ldots, m$, construct $\Psi_{H,t}$ and call the one fixed width-three counting algorithm. Theorem 3.3 gives at most $C(s+1)$ variables for an absolute C. The encoding lengths are polynomial in $L + 1$. Since $f_\alpha(C(s + 1)) = O(f_\alpha(s))$ for large s, each call has the asserted exponential bound. The $m + 1$ calls and the inversion of Lemma 3.1 add only polynomial overhead.

Make the calls sequentially, keeping only their integer outputs. The formulas have polynomial length, each call uses polynomial space, and all moment and transform values have polynomial bit length. The output has $m + 1$ coefficients of at most $n + 1$ bits each. Thus the whole algorithm uses polynomial space. □

Corollary 4.2 (Parameterized decision)**.** *Under* Count_α*, an instance* I *of* Set Splitting *with target* k *can be decided deterministically in polynomial space and time*

$$|I|^{O(1)} + 2^{O(f_\alpha(k))}. \tag{8}$$

In particular, Count_1 *gives* $2^{O(k \log\log k/\sqrt{\log k})} + |I|^{O(1)}$*, and* Count_2 *gives* $2^{O(k(\log\log k)^2/\sqrt{\log k})} + |I|^{O(1)}$*. Both are* $2^{o(k)} + |I|^{O(1)}$*.*

Proof. Trivial targets can be answered immediately. Apply the decision kernel of Lokshtanov and Saurabh [1, Theorem 1]. It either answers the instance or returns an equivalent instance (H', k_0) with at most k vertices and $2k$ edges. Keep k as the original parameter; no equality between k and k_0 is needed. Compute $P_{H'}$ by Theorem 4.1 and answer yes exactly when a coefficient at an index at least k_0 is positive.

The kernel has $s' \le 3k$ and a representation of polynomial bit length in $k + 1$. Its polynomial factors are absorbed in $2^{O(f_\alpha(k))}$ for large k. The initial reduction costs polynomial time in $|I|$. Since $(\log\log k)^\alpha/\sqrt{\log k} \to 0$, the dependence is subexponential in k. □

Remark 4.3 (A decision kernel is sufficient)**.** The computation in Corollary 4.2 counts colorings of the *reduced* instance. We do not assert that the kernel preserves the original coloring distribution. The full-profile theorem applies directly to the original input with parameter $n + m$, whereas the target-k corollary is a decision result.

Corollary 4.4 (Linear incidence)**.** *Assume* $\mathsf{Count}_{\mathrm{sp}}$*. For every fixed* B*, hypergraphs with* $J \le B(n + m + 1)$ *admit full-profile computation in polynomial space and time*

$$(L + 1)^{O(1)} 2^{O_B(s/\sqrt{\log s})}.$$

Proof. The selector formula has $v \ge n + m$ declared variables and $O_B(n + m + 1)$ clauses. Its density is bounded by a fixed constant. Apply $\mathsf{Count}_{\mathrm{sp}}$ and repeat the proof of Theorem 4.1. □

Why the sparse rate cannot be used for every kernel. The kernel bounds vertices and edges, not incidences. It permits $J = \Theta(k^2)$, so the selector formula may have $\Theta(k)$ clauses per variable. The general parameterized bound therefore uses Count_1 or Count_2, not $\mathsf{Count}_{\mathrm{sp}}$. Introducing incidence variables to make a circuit encoding sparse would sacrifice the linear variable bound.

5 Pinned colors and other predicate families

Proposition 5.1 (Pinned profiles and optimal colorings)**.** *Theorem 4.1 continues to hold when any subset of vertices has specified colors and the profile counts only extensions of these pins. Under the same premise, an optimal coloring of the original hypergraph can be found within the time bound (7) and polynomial space.*

Proof. Add unit clauses on the pinned color variables to every selector formula. This adds at most n clauses and no variables, and leaves the moment interpretation valid on the restricted set of colorings. Conflicting pins give the identically zero profile.

First find the largest exponent b with positive coefficient in P_H. Process vertices in order. Tentatively pin the next vertex to zero and compute the restricted coefficient at b. Keep zero if that coefficient is positive; otherwise pin it to one. The invariant that some optimum coloring extends the retained pins guarantees success. At most $n+1$ profile computations are used, adding a polynomial factor. This is a search result in $n+m$; it does not assume a target-k solution-lifting property of the decision kernel. □

The selector construction is not specific to equality predicates. The following formulation isolates its scope.

Theorem 5.2 (Families of 2-CNF predicates)**.** *Let $\Phi_1, \dots, \Phi_m$ be explicitly represented 2-CNF formulas on the same n declared Boolean variables, with C clauses in total. For $x \in \{0,1\}^n$, let $q(x) = |\{i : \Phi_i(x) = 1\}|$. The entire distribution of $q(x)$ reduces to $m+1$ exact 3-CNF counting calls, each with $O(n+m+1)$ variables and $O(C+m+1)$ clauses. Consequently Count_α gives time $\mathrm{poly}(L+1)2^{O(f_\alpha(n+m))}$ and polynomial space. A fixed bound $C = O(n+m)$ permits $\mathsf{Count}_{\mathrm{sp}}$ and its sharper rate.*

Proof. For every clause D in Φ_i, include the guarded clause $\neg y_i \vee D$, whose width is at most three. A selected predicate must be true; an unselected predicate imposes no condition. Add the same exact-t selector circuit as before. For fixed x, its count is $\binom{q(x)}{t}$. Lemma 3.1 recovers the distribution. An empty conjunction is true and imposes no guards; an empty clause produces the unit guard $\neg y_i$, correctly making that predicate unselectable. The size and space arguments are unchanged. □

Monochromaticity of a nonempty edge is the 2-CNF predicate

$$\bigwedge_{v \in e \setminus \{a\}} ((\neg x_a \vee x_v) \wedge (x_a \vee \neg x_v)),$$

so Theorem 3.3 is a direct specialization.

Corollary 5.3 (Arbitrary-width CNF satisfaction profiles)**.** *Under Count_α, the number of assignments satisfying exactly j clauses, for every j, of a CNF with n variables and m clauses of arbitrary width can be computed in $\mathrm{poly}(L+1)2^{O(f_\alpha(n+m))}$ time and polynomial space. An assignment maximizing the number of satisfied clauses can be found within the same asymptotic bound.*

Proof. For an original clause D_i, its falsification predicate is the conjunction of the negations of all its literals. This is a 1-CNF and hence a 2-CNF predicate. Apply Theorem 5.2 to the falsification predicates and reverse the resulting distribution. Pinning and self-reduction work as in Proposition 5.1. □

6 Relation to polynomial identities and earlier approaches

The transform $B_H(w) = Q_H(1 + w)$ is a standard binomial-moment identity. For nonempty hyperedges, White's coboundary polynomial [5] specializes to Q_H. For a nonempty selected edge e, let H/e delete that occurrence and identify its vertices, retaining every other edge label. White's Theorem 9, specialized to two primary colors and two colors in total, yields

$$Q_H(z) = Q_{H-e}(z) + (z-1)Q_{H/e}(z),$$

and therefore $P_H(z) = zP_{H-e}(z)+(1-z)P_{H/e}(z)$. We credit these identities as prior algebraic structure. The present reduction evaluates the coefficients of the shifted polynomial through bounded-width guarded formulas rather than traversing the deletion–contraction tree.

An alternative route processes every edge of size at least r by that recurrence. A root-to-leaf path has at most m decisions and at most $\lfloor (n-1)/(r-1) \rfloor$ contractions. Thus the number of leaves is at most

$$\sum_{j=0}^{\min\{m,\lfloor (n-1)/(r-1)\rfloor\}} \binom{m}{j}.$$

At a leaf a direct circuit encoding uses $O(n + rm)$ variables. With a choice of r balancing tree size against SAT cost, the sparse counting premise gives an exponent of order $s\sqrt{\log\log s}/(\log s)^{1/4}$. The selector reduction avoids both this branching overhead and the incidence-variable expansion. Even the more conservative Count_2 selector bound is asymptotically smaller than that exponent, since $(\log\log s)^{3/2}/(\log s)^{1/4} \to 0$.

The imported SAT manuscript already notes a bounded-rank, all-edges splittability consequence [3]. Our reduction handles unbounded rank, arbitrary targets, and exact profiles with a variable bound depending only on $n + m$. This distinction is the relevant transfer issue.

7 Verification, limitations, and open questions

The verification harnesses are `research/selector_moments.py` and `research/verify_paper.py`. They construct the guarded formulas, count them with a small exhaustive DPLL oracle, and check the inverse transform against exhaustive assignments. Verification passed on 378 hypergraph profiles using 1,773 moment counts, 20 optimal-coloring self-reductions, 80 general 2-CNF predicate profiles, and 20 CNF violation profiles. The manifest also records 27 encoding-size checks. The accompanying code is a check of the reduction, not an implementation of the imported fast counter. Appendix A specifies the reproducibility interface. Finite tests do not prove an asymptotic complexity claim; the size bounds, unique-extension argument, inversion, and bit analysis above supply the proof of the reduction.

The scope note and revised manuscript report different quantitative counting rates. We have not rechecked the entire external theorem or its Lean development. All corollaries are therefore stated against the interfaces that they actually use. No practical speedup is established here; hidden constants and polynomial degrees in an imported algorithm can dominate feasible instances.

A fuller literature audit is needed before claiming priority for the application. The coboundary polynomial, binomial inversion, cardinality circuits, and decision kernel are established ingredients. The central algorithmic point is that one-way guarded predicates retain linear *variable* complexity while permitting polynomially many clauses.

Two quantitative questions remain. First, can unrestricted-rank split profiles be reduced to fixed-density 3-CNF using only $O(n + m)$ variables? That would allow the sparse premise to remove the remaining $\log\log(n + m)$ factor. Second, can a sharper direct counting interface for these guarded formulas improve the general-density rate? The present proof does not answer either question.

References

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A Explicit reduction and output conventions

The following pseudocode describes the full-profile algorithm. The counter is an interface argument; supplying an exhaustive counter gives a small verification implementation, while the complexity theorem requires the stated subexponential interface.

```
SplitProfile(H, Count3):
  n := number of declared vertices; m := number of indexed edges
  for t := 0 to m:
    declare x[1..n], y[1..m]
    for each nonempty edge e[i]:
      choose anchor a in e[i]
      for v in e[i] except a:
        add (not y[i] or not x[a] or x[v])
        add (not y[i] or x[a] or not x[v])
    add unique-extension exact-t circuit on y[1..m]
    F[t] := Count3(resulting formula, all declared variables)
  for j := 0 to m:
    a[j] := sum over t=j..m of (-1)^(t-j)*binom(t,j)*F[t]
  return [a[m], a[m-1], ..., a[0]]
```

The return value is indexed by the number of *split* edges. All edges retain their occurrence labels, including duplicates. For $n = 0$, the one empty coloring still contributes one. Isolated vertices remain declared variables, so they contribute correctly to every count. Pinning adds units to the color variables before each counting call.

For independent reproduction, run

```
python3 research/verify_paper.py
```

from the repository root. This invokes the selector harness and writes `paper/verification.json`, including deterministic seeds, check counts, encoding sizes, and source hashes. The harness imports the constant-fan-in circuit and exhaustive counting utilities in `research.py`. It checks every simple hypergraph on at most three vertices, including empty and singleton edges, and seeded additional instances with parallel edges and isolated vertices. Its assertions verify the guarded count identity, binomial inversion, clause width, and a linear bound on the number of variables. Additional pinned-color and predicate-family checks test the extensions proved in Section 5. These finite checks are not premises of the mathematical proof.

B Source separation

For clarity, the proof dependencies are:

- Theorems 3.3 and 5.2, and Lemma 3.1, are elementary reductions with full proofs here.
- Theorem 4.1 and the profile/search corollaries use exactly the selected counting interface from Definition 2.1.
- Corollary 4.2 additionally uses the established decision kernel [1]; it does not assume a counting kernel.
- Corollary 4.4 uses the separate fixed-density interface. It is not substituted into the general kernel bound.